Lichen as built

the BABE gadget, the hub's ledger and the exit

draft

waxwing

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a companion to *Lichen*, version 0.4 [1]

Abstract

This companion to Lichen [1] gathers what is built and measured, in two parts. Part I is the verification gadget, BABE [6], which serves any L2. BABE's statement is fixed when the parties set up, while Alice chooses the amount b only when she withdraws; so the amount moves into the witness, together with the hub's one-time signature on it, which the hub must give on chain for exactly the amount Alice asserted there, or lose. BABE's cryptography is unchanged. We give the transaction graph as built, a tower that holds none of Alice's keys, the setup by verifiable secret sharing (19 seconds per contract on one laptop; the user keeps about 240 MiB), and the measurements: on Mutinynet with a real RISC Zero proof, a true claim challenged by the hub and answered took 30.5 kvB in five transactions, against BitVMX's 100–212 kvB in 23–25; unchallenged, 9.7 kvB in four; a false claim, 29.9 kvB in five. Part II is lichen's own L2, in which the hub is also the sequencer, so that the withdrawal statement must enforce what a separate L2 would vouch for: validity, by a chain of proofs; one history, final on Bitcoin, by a chain of single-use seals with the contract's work on top; and an exit from a state already anchored, against an operator that censors its users or vanishes, which the hub may answer only with a counter. An exit's dispute is about 15 kvB. The worst case falls on Alice: proving a forged counter false verifies a Groth16 proof inside RISC Zero, 56.3M cycles, about an hour on a laptop, and we set out the alternatives. The security argument is still a sketch, and data availability is the main open problem.

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1. Scope

The main paper [1] describes a contract between a user, Alice, and a hub: a 2-of-2 output holding her deposit, the hub's liquidity up to a cap $V_{\max}$, and two reserves. The hub's tokens on an L2 are collateralised by it; a withdrawal of b is a return of b tokens to the hub tied to the contract c, folded cooperatively or, if the hub refuses or is offline, enforced on Bitcoin by a verification gadget. The withdrawal statement is a predicate the L2 supplies, $\mathcal{R}_L(\mathcal{D}, c, \ldots; w)$, with $\mathcal{D}$ the deployment's parameters and w the witness; how the L2 makes its account of its ledger trustworthy is its own business. This companion gathers what lichen builds [3] on either side of that interface:

- *Part I, the gadget*: BABE, with the amount authenticated by the hub, as built and measured. It serves any $\mathcal{R}_L$.
- *Part II, lichen's own L2*: a ledger whose sequencer is the hub, and the statement that must then enforce validity, finality and an exit against it.

lichen's first gadget was BitVMX's interactive search [13], described and measured in its own companion [2], which is no longer maintained; it is the comparison throughout Part I.

What is ours is narrow in Part I: the binding of a late-chosen amount by the hub's authentication (Section 5), first put forward in a discussion on Delving Bitcoin [12]; the transaction graph and the setup by verifiable secret sharing, as built; and the measurements. BABE, its witness encryption and its garbled circuit are its authors' [6, 7, 8]; the problem of public inputs that are known only at proving time has been treated at the cryptographic layer by GOAT's researchers [11]. Part II is lichen's design throughout.

Figures marked (BABE) are the BABE paper's measurements, taken from its revised version (October 2026); figures marked (lichen) are measured in lichen's proof of concept, on regtest under mainnet's relay policy or on Mutinynet as stated; the rest are estimates, marked (est.). Where the design's estimates, made before the gadget was built, differ from the measurements, they are kept beside them. Proving times are on one 14-core Apple M4 Pro laptop, on its CPU (the RISC Zero version we use has no Metal prover), or on a rented 4-vCPU x86 server, as stated.

Part I

The gadget

2. BABE in brief

Groth16's verification, written additively in the target group, is

$$e(\pi_1, \pi_2) + e(\pi_3, \delta) = T, \qquad T = e(\alpha, \beta) + e(L, \gamma), \qquad L = IC_0 + \textstyle\sum_i s_i \, IC_i,$$

where π_1, π_2, π_3 are the proof, s_i the public inputs, and the rest the verifying key. Witness encryption (WE) lets a party encrypt a message so that only a holder of a valid proof can decrypt it. For pairing relations *linear* in the witness there is a simple scheme [7]: for a secret scalar r, publish r times the public elements; a holder of the witness pairs it against them and

obtains $r\,T$, which unmasks the message. Groth16's check is not linear in the witness, because of $e(\pi_1, \pi_2)$. BABE's idea is to supply $r\pi_1$ by two-party computation once π_1 is known:

- *Setup.* The verifier picks r and a message *msg*, publishes the WE ciphertext ($r\delta$ and *msg* masked by $r\,T$; 480 bytes), commits to *msg* by its hash, and garbles a circuit computing $r\pi_1$ from the bits of π_1, with r inside. The circuit is small because group addition to a fixed point is linear in a six-term feature vector of the point, so all but a point-validation gadget is a decomposable randomized encoding with no garbled tables, after Argo MAC [8]: 22 MiB per instance, 175 ms to make (BABE).
- *Assert.* The prover posts π_1 under a Lamport signature (508 bits).
- *ChallengeAssert.* To contest, the verifier must post the circuit's input labels for exactly those bits; a script checks that its labels and the prover's Lamport signature sign the same value. (A garbled circuit's input labels *are* a Lamport signature of its input.)
- The prover evaluates the circuit, obtains $r\pi_1$, completes the decryption with π_2 and π_3, and, if the proof is valid, publishes *msg* (*WronglyChallenged*), which blocks the verifier's *Failure*; after a timelock Δ_2 the prover's *Success* pays. If the verifier never challenges, Success pays after Δ_2 anyway. Failure needs Δ_1 after the challenge.

On chain, a contested valid proof costs Assert 9,240 vB, ChallengeAssert 17,400 vB and Wrongly-Challenged 149 vB (BABE), with one set of labels. Against a verifier who might garble wrongly (one whose circuit never lets the prover decrypt), the prover checks the garbling by cut-and-choose: for example 2,268 instances of which 4 are kept, 1 min 37 s and 1.8 GB at peak on a 16-thread laptop (BABE). Without more, the verifier must then post 4 sets of labels; *soldering*, a SNARK proving the kept instances' labels consistent, brings it back to one at about 12 minutes more of setup (BABE); verifiable secret sharing does the same in seconds, and is what lichen builds (Section 8). The WE is extractable, proved in the generic group model. The protocol is proved robust (an honest prover is paid) and knowledge-sound (a prover who is paid knows a witness) for a setup that both parties run honestly. Against a verifier who garbles wrongly, cut-and-choose is argued, failing with probability at most 2^{-40} for the parameters above, but not proved (BABE).

3. Where BABE fits in lichen

The proving stack is unchanged. lichen's withdrawal proof is a Groth16 proof over BN254: RISC Zero's recursion compresses the guest's STARK receipts into a succinct receipt, and RISC Zero's shrink-wrap proves that receipt's verification in Groth16, with RISC Zero's single trusted setup [15]. BABE verifies exactly such a proof. Any proofs the L2's statement composes, such as lichen's chain proofs, and the wrap stay as they are; BABE's withdrawal guest is the withdrawal statement with one more check and a different journal (Section 5).

The roles. Alice, who claims her withdrawal, is BABE's prover; the hub, the only party a false claim can rob, is its verifier, and so the garbler. BABE is a protocol between one prover and one verifier, both known at setup, which is what a lichen contract is: no committee, no open set of challengers, and no one else's funds in the output. Designs whose challengers are not known in advance pay heavily to approximate this; a bilateral contract has it for free.

Setup is per contract, and two-party. The hub garbles for each contract; Alice checks the garbling by cut-and-choose and keeps the finalized instances. This replaces nothing in lichen's trust model: as under BitVMX, the contract is set up by its two parties alone. What it adds is cost at opening (Section 9.5).

The two parties' conduct at setup protects different funds. A hub that garbles wrongly could refuse a valid withdrawal, since Alice could never decrypt; cut-and-choose is what rules this out, so Alice's funds rest on it. Alice's part in setup is small (her keys, her pre-signatures, and which instances to keep), and every secret is the hub's, so the hub's funds rest on the proving phase, where BABE's proofs apply.

One-shot. Revealing labels spends a garbled circuit's security, so a BABE instance settles one claim. A lichen contract has at most one withdrawal, after which it closes; renewal is a new contract with a new setup.

4. The fixed statement

BABE's ciphertext is built from T, and T contains $e(L, \gamma)$, the combination of the public inputs. So the whole public input is fixed at setup. For a RISC Zero proof that input is the claim digest, a hash over the image id and the journal; under BitVMX the journal is $b \,\|\, c \,\|\, \mathcal{D}$, so setup would fix b, which Alice chooses only when she withdraws. The ways round it that we know of:

- *Supply rL at proving time* for the asserted public input only, by envelopes opened with labels the verifier reveals per bit. With a single scalar public input this fails: the points $r\,2^j\,IC$ share one base, so one of them gives away $r\,IC$ and with it the ciphertext for any input. It needs each bit to be its own public input with an independent base, so a custom Groth16 wrap and its own trusted setup, and a garbled SHA-256 between b and RISC Zero's digest.
- *Deferred binding at the cryptographic layer*, as GOAT's researchers propose [11]: a blinding point and further garbled circuits for the dynamic part of the input, one of them about 200 million gates, and a larger challenge transaction.
- *One instance per digit value*: decompose b in base B and set up a fixed statement for each nonzero digit at each position. A dispute then proves each nonzero digit separately; base 16 for a 14-bit amount means 60 setups per contract and up to four disputes.
- *Refresh the statement* whenever Alice's balance changes, which needs contracts in channels so that the refresh is off chain, and a setup per change of balance.
- *Set up only when withdrawing.* Rejected: a hub that declined the setup would leave Alice's tokens without collateral, which breaks the contract's first property, that within its cap Alice trusts the hub for nothing.

We take none of these. Section 5 moves b into the witness instead, so one fixed statement, set up once per contract, serves any amount, and Script reads b from the parties' signatures.

5. Authenticated amounts

5.1. The statement

Let the hub hold, per contract, a one-time signing key pk_H, fixed at setup. The statement proved becomes

$$\exists\, b, w, \sigma_H : \quad \mathrm{Verify}(pk_H, b, \sigma_H) = 1 \ \wedge\ \mathcal{R}_L(\mathcal{D}, c; b, w) = 1,$$

where $\mathcal{R}_L$ is the L2's withdrawal statement unchanged (a return of b to the hub tied to the contract c; for lichen's own L2, the statement of Part II) and w its witness. The journal holds only values fixed when the contract opens: $c \,\|\, \mathcal{D} \,\|\, H(pk_H)$. The claim digest, and so BABE's statement, is then fixed at setup, while b is free until Alice withdraws.

5.2. The flow

1. *Request.* Alice asserts b on chain under her one-time key for it, as under BitVMX.
2. *Response.* The hub must sign the same b under pk_H, on chain, within its window w_H. The response's script verifies both signatures and checks, digit by digit, that they sign the same value. If the hub is silent, Alice's timeout pays her the asserted b, as the hub's silence does under BitVMX.
3. *Proof.* Alice proves the statement, using the hub's signature as part of her witness, and wraps the proof. Her wallet stops after the response with the hub's signature written out for the prover, and resumes when given the proof; she has until an absolute deadline T_A, shortly before T_{close} (Section 6).
4. *BABE.* Assert, ChallengeAssert, WronglyChallenged or Failure, as in Section 2.
5. *Payout.* Success pays through the bit outputs, which read b from Alice's signature, as under BitVMX.

5.3. Why the amount is bound

The hub's signature is not a judgement. It means "this is the amount being adjudicated", not "this withdrawal is valid", and an honest hub gives it mechanically for whatever Alice asserted, true or false. Three links bind the amount proved to the amount paid:

- the proof's b is the hub's, because the guest checks the hub's signature;
- the hub's b is Alice's on-chain b, because the response's script compares them digit by digit;
- Alice's b is the one paid, because the bit outputs read her signature, and a second signature from her under the same key forfeits them (`equiv_b`).

Suppose Alice asserts 100 but can prove a withdrawal of only 10. She has the hub's signature on 100 and no valid withdrawal of 100, or a valid withdrawal of 10 and no signature on 10. Neither satisfies the statement.

The authenticator must be the hub. Alice's own signature on b inside the witness would bind nothing: she could sign 10 privately for the proof and 100 on chain for the payout, and the

private signature, hidden in a zero-knowledge witness, would never be seen, so the equivocation could not be shown. A key she does not hold, used only on chain, is what carries the on-chain value into a fixed statement. The hub must therefore never sign a second value under pk_H, not even off chain; its key is used once, in the response, and the cooperative fold never touches it.

The payout could check both signatures. Since the hub's signature on b is on chain, each bit output could require both parties' signatures on the same value: Alice's leaf with bit i set, the hub's with it clear. Neither can then spend with a value the other did not sign, which removes `equiv_b` and Alice's wait δ at about the same script cost. This is not built: the bit outputs are BitVMX's.

5.4. The guest

The guest takes b and σ_H as private inputs, checks σ_H against pk_H, runs the withdrawal statement, and commits c, $\mathcal{D}$ and pk_H's digest. LN-GAP's one-time signatures chain `HASH160`, RIPEMD-160 of SHA-256, which has no accelerator in RISC Zero's zkVM; the hub's key instead chains SHA-256, which the zkVM accelerates and Script checks with `OP_SHA256`, at 32 bytes a chain value. Alice's key for b stays LN-GAP's, so the response's script verifies one signature of each kind. With lichen's statement (Section 14) the guest runs in 482k user cycles against the plain withdrawal guest's 333k, one segment either way (lichen); the design estimated the check at a few thousand cycles.

5.5. Security, as a sketch

The statement cannot say that the hub signed b, only that a valid hub signature on b exists, and one exists for every b; only the hub can compute it. So what binds the amount is not that a false statement cannot be proved but that the prover must *know* a witness: that whoever is paid knows one, which then contains a hub signature, which unforgeability then forces to be on the asserted amount. The pipeline already relies on this: lichen's chain proof's checks of users' signatures bind in the same way.

BABE gives the first step. Its knowledge soundness (Definition 9 of [6]) states that for any NP relation, a prover who gets the Success transaction onto the chain knows a witness, in the sense that an extractor recovers one from its view; it is proved for an honestly run setup (its Definition 16 and Theorem 8). Here the relation is the statement above, through two more layers: Groth16 is knowledge-sound (in the generic or algebraic group model), so the Groth16 proof yields a valid RISC Zero receipt; RISC Zero's STARKs are arguments of knowledge (in the random-oracle model), so the receipt yields the guest's input, (b, w, σ_H). If $b \neq b^*$, the asserted amount, then σ_H is a signature under pk_H on a message the hub never signed, and the hub signed exactly one: a forgery against a one-time signature.

What remains to be written properly:

- BABE proves knowledge soundness for an honestly run setup, not yet for one in which the prover takes part. Alice's part is small and every secret is the hub's (Section 3), so we expect the extension to be routine.
- The hub's signature arrives during the proving phase, after the statement is fixed, and depends on Alice's own assertion: the extraction must hold with the signature in the prover's

view, which BABE's definition allows but which should be checked against its proof.

- The composition of extractors across BABE, Groth16 and the STARKs, under BABE's restriction to benign auxiliary input.

6. The transaction graph

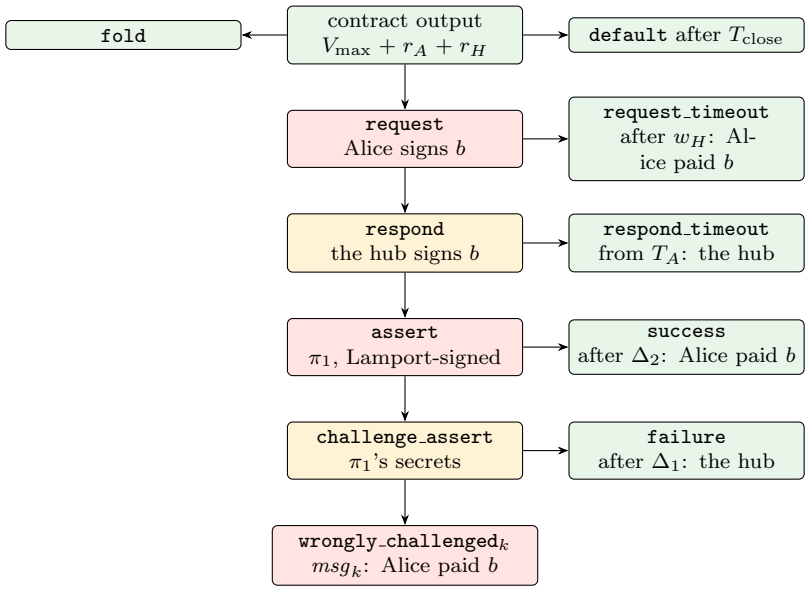


Figure 1: The dispute's graph as built. Red: Alice's links; yellow: the hub's; green: the terminals, each of which ends the dispute. One chain of links carries the contract's value, each link's output with its terminal; a terminal and the next link spend the same output, so only one of them confirms. Lichen's exit (Section 15.3) adds a claim to `request` and a counter to `respond`.

BABE's graph (its Appendix A.1) keeps the prover's deposit apart and has Failure spend three outputs, because there Assert starts from an outside coin. Here the dispute starts from the contract output, so the value flows through one chain and every transaction has one input (Figure 1, Table 1). A challenge and `success` spend the same output, so they cannot race, and BABE's condition $\Delta_2 > \Delta_1 + 2u$ is not needed. Each kept instance has its own msg_k, so there is one `wrongly_challenged` per kept instance. A terminal cannot pay into a pre-signed transaction, since its broadcaster's fee input changes its id, so every terminal that pays Alice carries the payout's bit outputs itself: `wrongly_challenged` is 0.82 kvB rather than BABE's 149 vB. The links carry a 330-sat anchor for their poster and pay a floor of their own, 1 sat/vB, out of the contract, so they relay alone.

All are pre-signed at opening: the default, four links, four terminals, and a `wrongly_challenged` for each kept instance, 19 transactions at $(N, M) = (78, 10)$ (the design estimated about ten); lichen's exit adds one. Each `wrongly_challenged` is signed by Alice alone and kept from the hub (Section 7). The payout and its bit outputs are as under BitVMX.

transaction	poster	spends, and what it carries or checks
`request`	Alice	the contract output; her one-time signature on b (one word)
`request_timeout`	Alice	`request`'s output after w_H: the hub never responded; the payout
`respond`	hub	`request`'s output, within w_H; the hub's signature on b, and Alice's again for the digit-by-digit comparison
`respond_timeout`	hub	`respond`'s output from the absolute T_A: Alice never asserted; the hub takes everything
`assert`	Alice	`respond`'s output, before T_A; π_1 under her Lamport key, in 2-bit digits, with π_2 and π_3 beside
`success`	Alice	`assert`'s output after Δ_2: no challenge; the payout
`challenge_assert`	hub	`assert`'s output, within Δ_2; the secrets of π_1's digits, by completing Alice's adaptor pre-signatures (Section 8)
`wrongly_challenged`$_k$	Alice	`challenge_assert`'s output, before `failure`; msg_k of kept instance k; the payout
`failure`	hub	`challenge_assert`'s output after Δ_1: no answer; the hub takes everything

Table 1: The transactions of the dispute, as built.

The assertion's deadline. Alice's heavy move, `assert`, needs no window: a late assertion only delays a payout the hub owes anyway, and the hub loses nothing by waiting. So its deadline T_A is absolute, T_{close} less Δ_2, Δ_1 and δ, so that the rest of the dispute and the payout fit, and the proof, which needs the hub's signature first, never races a timeout. On Mutinynet, with 30-second blocks, the windows are $w_H = 40$ (240 with the exit, Section 16), $\Delta_2 = 240$, $\Delta_1 = 80$ and $\delta = 40$ blocks, and a test contract lasts a week, so an assertion is due a week after the contract opens. What remains due within a window is, for Alice, `wrongly_challenged` and, for the hub, its response and its challenge.

7. Alice's liveness: a keyless tower

Under BitVMX, Alice cannot delegate her moves without her keys. Each move signs, with a one-time key, hashes from her trace at positions the hub's last choice selects, so it cannot be signed in advance; a tower that plays her side holds her move keys, and by equivocating could lose her the contract [2, §10]. Making those moves keyless would need `OP_CAT`. BABE removes the problem rather than shrinking it: Alice's only commitments, b and π_1, are fixed before anything she must answer, and her one answer to the hub is a hash preimage, not a signature. This is a difference of kind, not of count: a gadget with fewer interactive moves would still need someone holding her keys while the dispute runs; here no one does.

What she signs, and when. Two one-time signatures: on b, for `request`, made when she returns b on the L2; and on π_1, the Lamport signature of `assert`, made once she has the proof. The proof needs the hub's signature on b, so the second comes after `respond`, but before the loose, absolute deadline T_A: a second visit, not a race. A prover she does not trust with keys can make the proof; she checks it (one Groth16 verification) and signs π_1.

What the tower holds. Her signed `request`, her Lamport signature on π_1 (a preimage per digit, checked against her key), her signatures on the `wrongly_challenged`s, the proof, the contract's pre-signed transactions and her adaptor pre-signatures, the kept instances and the opened shares (about 240 MiB at $(78, 10)$), and the key of her fee budget, which pays the terminals' fees. From these and the chain it plays everything left to Alice:

- `assert`, from her signature on π_1 (`request` and `respond` are on chain already, since the proof needs the hub's signature);
- `request_timeout` and `success`: pre-signed, released by their timelocks;
- `wrongly_challenged`: it reads the secrets from `challenge_assert`, interpolates each kept instance's labels, evaluates the garbled circuits to $r\pi_1$, decrypts with π_2 and π_3, and posts the *msg* of the instance that opens.

Her bit outputs wait δ for her own key, as under BitVMX, and nothing races her for them: she sweeps them later from her own device, with the tower's state file.

On Mutinynet a tower played contract 1001's dispute this way, from lichen's Docker image on a rented 4-vCPU machine [4]: Alice requested, proved after the hub's response and exported the package (740 KB, and 243 MB of kept instances); the tower asserted and, the hub not challenging, posted `success` the block after Δ_2. Her device did nothing between the export and the sweep.

Two changes to BABE's graph, made. First, in BABE, the leaf `wrongly_challenged` spends checks the prover's signature as well as the hashlock, $\mathsf{HashLock}(h_{msg}) \wedge \mathsf{CheckSig}(pk_P)$, signed when posted (its Appendix A.1). A tower cannot sign then, but the signature should stay. Any spend of the leaf reveals *msg* and blocks `failure`, which is what Alice wants, but only once it confirms, and the hub knows *msg* from the start. Without the signature the hub could spend the leaf itself, in a large transaction at a low fee rate, which Alice's spend could replace only by paying more in all (BIP 125's third rule); if the hub's spend does not confirm within Δ_1, it replaces it with `failure`. So Alice makes the signature at setup instead: `wrongly_challenged` is fixed by then, and *msg* is in its witness, not under its signature. As every leaf in lichen's graph, it is 2-of-2, and she keeps her signature from the hub, for her towers only. She signs it `ALL|ANYONECANPAY`, as a TRUC (version 3) transaction, and the tower adds *msg* and a fee input of its own. A TRUC transaction cannot spend an unconfirmed parent of another version, so the tower's fee coin confirms first; `challenge_assert` has confirmed already, since Alice answers only a confirmed challenge. Bitcoin Core relays TRUC from version 28; on Mutinynet a node of an older build refused one with `version` while its newer peers relayed it, so lichen's wallet now sends a transaction its node refuses for its version through the network's public Esplora.

Second, fees. The terminals (`request_timeout`, `success`, and `wrongly_challenged`) are signed `ALL|ANYONECANPAY`, and their broadcaster adds a coin of exactly the fee from Alice's fee budget, set aside when the contract opened [2, §6]: the tower holds that budget's key, which guards nothing else. (At 1 sat/vB a small terminal's fee is under the dust floor of a P2TR output, so the coin is at least 330 sats.) The links (`request`, `assert`) pay their floor themselves; above it they are bumped through an anchor only their poster can spend, which lichen still locks to the key that also sweeps her winnings. So a tower posts her links at their floor, which sufficed on Mutinynet; to bump them it needs the anchors to have a key of their own, guarding nothing but

330 sats each, which is not yet built.

A compromised tower can do little. It holds no key that signs a move, so it cannot sign a wrong move or a second value. That matters here: given her Lamport key, a tower that signed two values of π_1 would let the hub combine their bits into a third, signed digit by digit, and challenge with the secrets for it; she could not decrypt, and `failure` would take everything. So the tower is not trusted like her own machine, as under BitVMX, but like a Lightning watchtower: several can run at once, strangers to her and to each other, and any one suffices. At worst one stays silent, or spends her fee budget, 30,000 sats on Mutinynet. One caveat: a tower run by the hub holds the signed `wrongly_challenged`, and could post it with *msg* as soon as it posts `challenge_assert`, padded with inputs of its own, to pin it as above. TRUC's limit of 10 kvB on the transaction bounds what an honest tower must outbid, about 100,000 sats at 10 sat/vB. Her wallet, for its part, signs one π_1 only, and keeps the proof it signed.

8. Setup by verifiable secret sharing

Fourteen minutes of computation per contract, paid again at every renewal, is probably more than opening a contract can bear; twenty seconds is not. Glock [9] replaces soldering with verifiable secret sharing (VSSS), and BABE's authors have prototyped that layer for BABE's 508 input bits (BABE, §9.4), without a transaction design. lichen builds it, with the transaction design below.

Why BABE keeps 4 of 2,268. Cut-and-choose fails only if every kept instance is bad and every opened one good, so the error is $\binom{N}{M}^{-1}$, held to 2^{-40}. Unsoldered, each kept instance puts a label set on chain; soldered, the proof's time grows with M (745, 866 and 969 s for 4, 5 and 6, BABE). Either way M must be small, and a small M forces a large N.

Secret sharing reverses that. The hub shares each input secret, for each digit and each value, across the N instances, with threshold $N - M + 1$, and publishes commitments to the polynomials (Feldman's, over secp256k1). Alice checks the opened instances' shares against them. The labels of a free-XOR garbling are related ($L^1 = L^0 \oplus \Delta$), so they cannot themselves be independent shares: the shared values are secrets of their own, which each instance maps to its labels through a small table, $H(s) \oplus L$. At proving time one secret per digit goes on chain, $f(0)$ of its polynomial; with the $N - M$ opened shares, Alice interpolates the secrets, and so the labels, of every kept instance. The value Alice did not assert stays hidden, since only $N - M$ points of its polynomial are known. The chain carries one secret set whatever M is, and the cost moves to N, the shares to commit and check. So the right choice is now a small N:

(N, M)	setup, both roles	Alice stores and receives
(2,268, 4), soldered	∼14 min (BABE)	∼89 MiB
(78, 10), VSSS	19 s (lichen)	∼240 MiB (lichen)
(124, 8), VSSS	∼1 min (BABE's estimate for the layer: 54 s)	∼190 MiB (est.)
(181, 7), VSSS	∼1.5 min (84 s for the layer)	∼170 MiB (est.)

$\binom{78}{10}^{-1}$ is $2^{-40.2}$; lichen's wallet refuses a setup of less than 2^{-40} off regtest. The bill moves to storage and transfer: Alice keeps every finalized instance, since she cannot know which one is honest. Glock's even split, about 44 instances with 22 kept, would be the wrong end of that trade here, at over 500 MiB.

On chain: adaptor signatures. The secret posted on chain must be the one Alice checked against the commitments, and the chain must check that. A hashlock cannot: tying a hash to a committed point sG needs a proof, which is what soldering is. Glock uses adaptor signatures instead. At setup Alice gives the hub, for each digit j and value β, a pre-signature on `challenge_assert` under her own one-time key $A_{j,\beta}$, adapted to the committed point $C_0 = f_{j,\beta}(0)\,G$. The script selects $A_{j,\beta}$ by her Lamport signature, which the hub copies from `assert`, and the hub can satisfy it only by completing her pre-signature, which hands her $f_{j,\beta}(0)$. The selection matters both ways: the other value's secret would leave Alice unable to decrypt, and both values' would let her evaluate the circuit on points other than π_1. In Glock the garbler is also the prover, so its adaptor signatures are its assertion; in BABE the prover asserts and the verifier reveals, so both are paid for. A completion for a value other than the asserted one does not verify (lichen).

Each secret is the adaptor of exactly one pre-signature, under a key used for nothing else. Nothing is adapted twice to one point, so the setting is adaptor signatures' basic one-time case; one secret serves all M kept instances, but by interpolation off chain, not through M signatures. The pre-signatures are all on one sighash, under different keys, which BIP340's challenge includes. Alice makes them once the kept instances are chosen: their hashlocks are in `challenge_assert`'s output, and so in the sighash.

Two bits a step. Bit by bit, this does not fit in one input. Tapscript limits an input's initial stack to 1,000 elements (BIP 342), and each bit needs two, the hub's signature and Alice's preimage: 1,016 for π_1's 508 bits, 1,017 with the hub's own signature, which lichen's tests confirm fails where 1,000 passes. Alice therefore signs π_1 in 2-bit digits, revealing one of four independent preimages per digit: Lamport over a larger alphabet, which, unlike Winternitz's chains, needs no checksum. The hub shares one secret per digit and value, which each instance maps to both bits' labels. The polynomials number the same, $254 \times 4 = 508 \times 2$, as do the keys and hashes in the leaf, but the hub's signatures and Alice's preimages halve, to 254 each, and the stack holds 509 elements:

	bit by bit (est.)	2-bit digits (lichen)
`challenge_assert`'s stack	1,017 elements: two inputs	509
`challenge_assert`	∼26 kvB	20.72 kvB
`assert`	9.24 kvB (BABE)	7.73 kvB
a false claim disproved	∼37 kvB	29.9 kvB
a true claim challenged to the end	∼39 kvB	30.5 kvB
the hub's windowed sum	∼27 kvB	21.5 kvB
contracts for a week's dispute	37,000	47,000

Wider digits do not pay: at three bits the keys and hashes per bit rise from two to 8/3. The

pre-signatures, 1,016, are a small transfer, about 100 KB.

The keyless tower survives. Alice's keys $A_{j,\beta}$ are used once, at setup, to make the pre-signatures, and never again. To read the secrets the tower needs her pre-signatures and the opened shares, none of them a key, and it extracts and interpolates as Alice would (Section 7).

9. Costs, measured

The sizes and disputes below were measured on Mutinynet before the graph carried lichen's exit; Section 16 gives what it adds.

9.1. Transactions

Table 2 gives the sizes as mined on Mutinynet (lichen), or on regtest under mainnet's relay policy where marked, beside the design's estimates.

transaction	measured	estimated
`request`: one signed word	0.45 kvB	∼0.4
`respond`: two signed words and the comparison	0.76 kvB	∼0.6
`assert`: π_1 in 254 2-bit digits	7.73 kvB	∼7.8
`challenge_assert`: 254 secrets by adaptor signatures	20.72 kvB (its leaf 60,232 bytes, 509 witness elements, regtest)	∼21
`wrongly_challenged`, with its fee input and the payout	0.82 kvB	∼0.21
`success`, with its fee input and the payout	0.78 kvB	∼0.3–0.8
`failure`, with its fee input	0.24 kvB	∼0.3–0.8
the hub's sweep of its bits; Alice's of hers	2.7; 1.7–1.8 kvB	∼3

Table 2: Transaction sizes, measured and as the design estimated them.

9.2. Disputes

Three disputes on Mutinynet (2026-10-29), each a withdrawal from a contract of cap $2^{21}-1$ sats, with a real proof [4] (Table 3). Every link went at its floor, 1 sat/vB, with no child: a dispute's fees are about its size in sats. Table 4 sets them against BitVMX's. Most of the gain against a vanished hub comes from not publishing the proof's input words before the hub moves; BitVMX could recover much of it by demanding them only from a hub that is present [2]. A false claim's dispute has one shape, whatever the lie: under BitVMX the liar chooses the lie, and with it the cost of the search that catches it, up to twice the cheapest.

ending	the dispute	what happened
a true claim, challenged	30.5 kvB in 5	the hub, hostile, challenged a valid proof; kept instance 9 of 10 opened; answered five blocks after the challenge
a true claim, unchallenged	9.7 kvB in 4	played by a tower in Docker; `success` the block after Δ_2
a false claim	29.9 kvB in 5	another contract's real proof; challenged, none of 10 kept instances opened; `failure`

Table 3: Three disputes on Mutinynet, 2026-10-29.

outcome	BABE	BitVMX [2]
the hub present: a fold	0.2 kvB	0.2 kvB
the hub gone	∼1.2 kvB, `request` and its timeout (est. from the measured sizes; played on regtest)	23.1 kvB
the hub refuses a true claim	9.7 kvB in 4; 30.5 in 5 if it challenges	∼120 kvB in 23
the hub disproves a false claim	29.9 kvB in 5	100.4 kvB
the worst case	the same: there is no separate case for reads	200–212 kvB, a read lie
Alice's transactions, those due in a window	4 (request, assert, wrongly challenged, success), one	13, twelve
keys Alice's tower must hold	none that signs a move: the fee budget's	her move and hop keys, and her input words'
pre-signed per contract	19 (20 with the exit)	141

Table 4: BABE's outcomes, as measured, against BitVMX's [2].

9.3. Congestion

The main paper's rule, $Ns \leq w \cdot 1\,\mathrm{MvB}$ for N contracts whose move of s bytes is due in a window of w blocks, must hold for every window. Comparing gadgets by one window would be unfair to the one with fewer: a BitVMX dispute is twelve windowed moves on each side, one after another, so a hub safe at a given window under BitVMX asks for a dispute twelve windows long. The fair basis is the dispute's length T for a given N, with each window sized to its move: $N \leq T \cdot 1\,\mathrm{MvB}/\sum_i s_i$, summed over the defender's windowed moves [2, §9]. Table 5 applies it to the measured sizes. Hostile users bind under BABE. For a dispute of the same length, BABE

	BABE, Alice against a hostile hub	BABE, the hub against hostile users	BitVMX, the binding side
windowed moves	`wrongly_challenged`, 0.82 kvB	`respond` 0.76 and `challenge_assert` 20.72	as built, Alice: twelve windows sized to her proof; with a window per move, the hub
$\sum_i s_i$	0.82 kvB	21.5 kvB	302 kvB; 121 kvB
contracts safe with a dispute of a day	∼175,000	6,700	480; 1,190
of a week	—	47,000	3,300; 8,350
the dispute for 60,000 contracts	49 blocks	8.9 days	126 days; 50 days

Table 5: Contracts a hub can hold before congesting the chain pays, by the dispute's length.

holds about 5.6 times as many contracts as BitVMX with a window per move, and 14 times as many as BitVMX as built. A hostile hub stops mattering, since Alice's only windowed move is 0.82 kvB. The hub's response, 0.76 kvB in w_H, is part of its sum: 60,000 contracts need w_H of at least 46 blocks. Chunking BitVMX's path into separate windows would not change these figures, since it shrinks the path's largest move but not its sum.

9.4. Reserves and time

A reserve must cover the other side's fees in a dispute it loses. The links pay their floors out of the contract, and the loser's reserve bears them: at most 30,981 sats at 1 sat/vB, the four links' floors and their anchors, against 197,356 under BitVMX (lichen). lichen's hub sizes its reserve to that figure, at the least. A hub that refuses a true claim and does not challenge loses only the first three links', 9,934; one that challenges pays for its own challenge too. At 10 sat/vB the deepest dispute's floors are about 310,000 sats, against about 1.2 million under BitVMX, so the contract suits balances from about a hundred thousand sats rather than half a million.

A hostile hub stretches a dispute to about $w_H + \Delta_2$ blocks after Alice's assertion, plus Δ_1 if it challenges. On Mutinynet the challenged true claim was answered 20 minutes after Alice requested, 15 of them her proving; the unchallenged one paid two hours after its assertion, that being Δ_2. A vanished hub's users wait w_H, as under BitVMX.

9.5. Setup per contract

The new cost, at lichen's $(N, M) = (78, 10)$ with verifiable secret sharing (Section 8), measured on one 14-core laptop (lichen), beside BABE's operating point with soldering (Table 6). Our

item	$(78, 10)$, VSSS (lichen)	$(2{,}268, 4)$, soldered (BABE)
an instance	23.83 MiB (gates 8.06, the encoding's tables 15.77); garbled in 1.2 s, evaluated in 0.6 s, one thread	22 MiB; 175 ms, 127 ms
the hub's setup	dealing 1.3 s, 78 instances 9.4 s	cut-and-choose 1 min 37 s for both roles, 1.8 GB at peak; soldering 745 s more
Alice's checks	1.5 + 8.1 s	(in the above)
in all	19 s (estimated 40)	∼14 min
what Alice downloads and stores	∼240 MiB, 10 instances	∼89 MiB, 4 instances
a hub opening 10,000 contracts	∼30 laptop-hours of the hub's own work	∼2,300 laptop-hours

Table 6: The setup per contract.

garbling hashes with BLAKE3 and multiplies schoolbook, where BABE's uses fixed-key AES and Karatsuba, hence the larger and slower instance. On Mutinynet the hub's setup took 9.5 s. The download is now the larger cost: about 240 MiB per contract, again at each renewal. Two ways to cut it remain open: Duty-Free Bits' 45-fold smaller encoding [10], unverified for these instances, and reusing the kept instances across renewals, sending a fresh *msg*, ciphertext and hashlock for each, since an unchallenged instance has revealed nothing.

Part II

Lichen's own L2

10. When the hub is the sequencer

At the main paper's level, which we call P_0, the L2's account of its ledger is trusted: in its simplest form, $\mathcal{R}_L$ checks a sequencer's signature on a state root and a Merkle path to the return. That is sound when the L2 is not the hub's, as for the BitVM bridges [14]: an L2 secured by its own consensus, or a rollup on another chain with its own proofs, supplies the account and lichen consumes it. The hub is then only a counterparty; it cannot rewrite a ledger it does not run.

In the L2 that lichen builds [3], the hub and the sequencer are one operator, or two that trust each other. That is a design choice, and its consequence is that the L2's account is the hub's own word. A hub that profits from defaults, since whatever isn't withdrawn by T_{close} is its, has every reason to bend that account. So everything a separate L2 would vouch for has to be enforced

inside lichen's $\mathcal{R}_L$, against the operator. We build it in two more levels:

- P_1, *validity and finality*: the return is in a state that ends a valid history from genesis, that history is the one anchored on Bitcoin, and the anchor has the contract's work on top (Sections 12–14).
- P_2, *the exit*: Alice can leave from a state already anchored, without a return, against an operator that censors her return, stops sealing or vanishes; the hub may answer only with a counter showing that she spent what she claims (Sections 15–17).

The gadget of Part I does not change between the levels; the statement does, and from P_2 on, so do the first two transactions of the dispute, which carry the claim and the counter.

With the hub and the sequencer one party, the threat is the operator against its users. Users against users is the ledger's ordinary job, and the operator against itself is out of scope. Table 7 lists what the operator could do with an unchecked account, and what answers each. A vanished operator needs no new mechanism: a hub that never responds to a `request` lets

the operator	answered by
mints notes nobody paid for, and withdraws them from users' contracts	validity: the chain proof (Section 12)
signs two valid histories that split, paying Bob in one and returning the same money to the hub in the other	one anchored history: the seal chain (Section 13)
anchors a history on a branch of Bitcoin it mined	the contract's work W on the anchor (Section 13)
censors Alice's return or stops sealing, so that she has nothing to prove, and takes her contract when she can't	the exit (Section 15)
withholds the data of something it anchored	not yet answered: data availability (Section 18.1)

Table 7: What a merged operator could do, and what answers it.

`request_timeout` pay Alice after its window, whatever she asserted (Section 6). The exit is for one that is present but hostile: it responds, then censors her return or stops producing the chain receipts she needs, so that she can't prove by the deadline and the contract is taken from her.

11. The ledger

The ledger is one of notes. A note is an owner, a value, a memo and a random ρ; its commitment is a SHA-256 hash of these, and the commitments form an append-only Merkle tree, so a note's path under a root stays valid however the ledger grows. A spent note is not removed but recorded by its *nullifier* in a second tree, a sparse Merkle tree keyed by nullifier, whose leaf records how the note was spent:

- *spent*: an ordinary transaction;
- *returned*(c, r): a transaction paying the hub exactly one note, of r with memo $c \neq 0$, the first

step of a withdrawal from contract c;

- *exited*(c): contract c's exit record (Section 15).

The tags matter only to the exit, whose counter must tell which spends may be held against which claim. A state root SR hashes the height, the commitment tree's root and the nullifier tree's root, and the sequencer signs it (BIP340).

There are two kinds of note in one tree. *Transparent* notes have a public owner key and value: the hub's mints (a deposit's tokens) and returns to the hub are transparent, and a transparent spend is authorised by the owner's signature. *Shielded* notes publish only their commitment; the owner's authority is knowledge of a key nk, the nullifier is derived from it so that nobody can link it to its note, and the sender proves the spend in zero knowledge on her own machine [5]. The transaction types are a mint, a transfer, shielding, a return, an anchor record (Section 13) and an exit record (Section 15).

Three proofs, all RISC Zero, are involved (Table 8).

proof	made by, when	shows; cost
spend	the sender, for each shielded transaction	its notes exist, are hers and are unspent, and the amounts balance; 6–30 s on an 8-core Mac (lichen)
chain	the sequencer's prover, for each sealed height	this block is a valid transition, and so was the history before it (each proof assumes the last); about 40 s for a small height, 142k cycles, on the laptop in Docker (lichen)
withdrawal	the user, or anyone she delegates to, only in a dispute	her statement, P_1 or P_2 (Sections 14 and 15); wrapped in Groth16 for the gadget

Table 8: The three proofs.

12. Validity

The chain guest proves one sealed height: the block's transactions applied to the previous state give the new one, its spends authorised (a signature, or a verified spend proof for a shielded note), no nullifier twice, values balanced, mints signed by the hub. It takes the previous height's chain receipt as an assumption, so each receipt vouches for the whole history from genesis. Its journal carries the height, the state root, a window of recent roots, the anchoring state of Section 13, and a history hash

$$hist_0 = 0, \qquad hist_h = H(\texttt{"lichen/hist"} \,\|\, hist_{h-1} \,\|\, SR_h),$$

which commits the whole sequence of roots to h, so that an earlier state's place in a later one's history is a hash-forward check.

Costs (lichen): a block of one mint is 395,543 cycles, a block holding only an anchor record about 130,000, both one segment; on the laptop a small height proves in about 40 s. The prover runs behind the seals, so for an honest sequencer the lag is only a delay, but every withdrawal waits

for the chain receipt at its return's height. One machine proving height after height keeps up with on the order of a hundred transactions an hour; beyond that the remedy is the operator's: a GPU, or proving blocks in parallel and chaining them afterwards.

A valid history is not a unique one. The sequencer can sign two histories that split at a height, each valid, which the chain proof cannot see. The wallet and the hub record every root they are shown and refuse to go on when shown a second root at a height, with the evidence; that makes equivocation visible to whoever looks, and anchoring makes it useless.

13. Finality

The seal chain. The deployment's genesis names an outpoint, the first *seal*, a P2TR output of the sequencer's key. An anchor spends the current seal and creates the next, beside an output `OP_RETURN "lichen"` $\|$ h $\|$ $hist_h$, committing the history to a sealed height h; it is 269 vB (lichen). Bitcoin lets an output be spent once, so there is one chain of anchors, after Todd's single-use seals [17]. The sequencer anchors at most once a Bitcoin block, and only when something has changed.

Anchors inside the history. A linear chain of seals could still commit roots of two histories in turn. So once an anchor confirms, the sequencer records it on the L2, and the chain guest checks the record: the transaction is in the block it names, by Merkle path to the header; it spends the seal the journal names, whose script is checked so that no one else could spend it and stall the anchoring; and it commits a *hist* that is an ancestor of the current one. A history forked from the anchored one can then hold no anchor after the fork: the only spend of the seal commits the other history.

The contract's work. The chain guest checks no proof of work: an anchor on a branch of Bitcoin costs the work on top of it, whatever the guest checked. The withdrawal statement checks it instead: headers that follow the anchor's block, each linked and within its own target, whose work sums to at least W. W is fixed when the contract opens, ten blocks' work at the difficulty of that moment (its checkpoint, which both parties check against their own view of the chain), and it goes into the contract's identifier,

$$\mathcal{D} = H(\texttt{"lichen/contract"} \,\|\, \text{deployment} \,\|\, \text{genesis seal} \,\|\, W),$$

the deployment being the sequencer's and the hub's keys. A forger gains nothing from headers declaring low difficulty: work is summed, so headers worth W cost about W hashes however it is split. About ten headers cost the statement a few thousand cycles. This is not a proof of the heaviest chain; it makes a forgery a matter of mining a fork of Bitcoin as long as W, which bounds the hub's exposure to one forged history by the reward of the blocks a fork replaces.

For receivers. A payment is final once its height is anchored ten blocks deep. The wallet does not take the sequencer's word for the anchors: it computes *hist* from the roots it has checked and finds each anchor on Bitcoin itself, so its balance splits into final, anchored and sealed only.

14. The withdrawal statement

At P_1, lichen's $\mathcal{R}_L$ proves:

1. a note of b, owned by the hub with memo c, marked a return, in the commitment tree of a state root SR, by Merkle path;
2. SR is the sequencer's, by its signature at its height;
3. a chain receipt, at that height or any later one, whose anchoring state commits SR: validity from genesis, in the one anchored history;
4. headers on the anchor's block with work at least W.

Under BABE the statement also checks the hub's one-time signature on b, and its journal is $c \,\|\, \mathcal{D} \,\|\, H(pk_H)$, the same for every amount (Section 5). The chain guest's image id is built into the withdrawal guest, so the one image id the contract pins covers both. The statement is 333k user cycles, 482k with the hub's signature, one segment either way (lichen). On the server the first proves in 359 s and the second in 640–660 s, a larger segment; on the laptop, about 30 s and a minute (est., from measured segments of the same sizes). The wrap takes about 255 s on the server and 134 s on the laptop, in RISC Zero's Docker image (lichen).

The cooperative path. The hub checks the same statement natively, in milliseconds, against the chain receipt and its own node's headers, and folds: one transaction of 193 vB. So it pays out only what a dispute could win, and Alice proves nothing. Anchoring adds a wait, not a cost: the next anchor and W on it, about two hours on mainnet.

15. The exit

A withdrawal starts with a return, which the sequencer must include. A merged operator profits from not including it. The exit lets Alice leave without one:

> *these notes of mine were unspent at the anchored state S; pay me b.*

The operator may answer once, only with a *counter*: something only Alice could have done, showing she no longer holds what she claims. It can't stop an exit otherwise: a censoring operator has no counter to a true claim, and a vanished one times out.

15.1. The claim

Alice claims from her *exit kit*, which the sequencer serves after each of her spends and receipts and her wallet checks against the signed root: at an anchored state S, each note's path in the commitment tree and its nullifier's absence from the nullifier tree. A kit never goes stale: the tree only grows, and absence from S's set stays true of S. If the operator stops serving, her last kit still works: if she hasn't spent since, no counter exists. A payment is final for its receiver once the receiver holds its kit.

The claim's fingerprint is

$$E = H(c \,\|\, O_D \,\|\, b \,\|\, h_S \,\|\, SR_S \,\|\, \mathit{nullifiers}),$$

O_D being the contract's deposit coin, which only its funding can spend, so that it names the contract on Bitcoin and is known when the setup fixes the statement. Each claimed note's owner signs E, and `request` posts the claim: h_S, SR_S, each note's opening and its owner's signature,

at most 1,169 bytes for eight notes, in fifteen witness items of up to 80 bytes, since Bitcoin Core relays no larger tapscript witness item. The statements read it through the coinbase's witness commitment. It is posted so that the hub knows which nullifiers to search for, and so that the signatures can be cited against her. A claim spends whole notes and the contract pays once, so the wallet claims everything it holds, up to $V_{\max}$; value above b in the claimed notes is forfeit, and the wallet sizes its notes to fit.

15.2. The counter

A counter shows one of the claimed notes gone, on one of two grounds.

A spend. The note's nullifier is in the nullifier tree of an anchored state S' after S, in the same history (S''s chain receipt hashes forward through S), with a leaf that counts:

- *spent* counts if S''s anchor confirmed in or before `request`'s block. A later spend could be a transaction of hers that the operator held and sealed after she committed to her claim; the cutoff leaves only what was final on Bitcoin when she claimed. `request` must have the contract's work on it too, or either side could forge the cutoff with a cheap header;
- *returned*(c', r) counts the same way, unless $c' = c$ and $r \geq b$: then the hub holds at least b of her tokens for this contract, which pays once. A smaller return, or one to another contract, can't shelter a spend;
- *exited*(c') counts whenever anchored, unless $c' = c$: a record of the same notes claimed through another contract.

An equivocation. The owner of one of the notes signed a claim of it through another contract. Under BABE the hub responds before Alice proves, so two requests through two contracts in one block would both fall due before either exit record existed; her signature on the other claim is the evidence instead. Only the owner can sign one, a copied signature names its own contract, and an honest wallet never signs a claim of a note through two contracts. No L2 state is involved.

Why the operator can't forge one. A nullifier enters the set only by a spend, which needs the note's keys, or by an *exit record*, which the hub posts once a `request` confirms, to retire the claimed notes, and which the chain guest accepts only with the owners' signatures on that claim. So every counter traces to something the owner did. Her own return to this contract never counts, so a return she signed, which the operator still holds, can't be sealed after her claim and used against her.

15.3. On the gadget

The claim and the counter ride on the dispute's first two transactions; the rest of the graph is Part I's (Section 6).

- `request` signs, beside b, the claim's E under a second one-time key of Alice's, and posts the claim. A return's dispute signs a fixed E and posts nothing, so one graph serves both.
- `respond` has two leaves, each pre-signed at opening and spending into one transaction, so that `assert` follows either, and both comparing the hub's b and E with Alice's digit by digit:

plain, the hub signs E under a key e_{plain}; *countering*, it signs E under e_{counter} and the counter K itself.

K is a Groth16 proof of the counter guest, whose journal is $\mathcal{D} \,\|\, E$, compressed to 128 bytes and signed whole, in 259 digits. It is signed rather than hashed because no script can tie a hash of posted bytes to signed digits without `OP_CAT`: a hub that signed $H(K)$ could post other bytes and leave Alice nothing to prove false. Uncompressed, at 256 bytes, it would exceed the 1,000-element stack limit at two elements a digit.

The statement's three branches. Alice's statement is fixed at setup, with the journal $c \,\|\, \mathcal{D} \,\|\, H(pk_H \,\|\, O_D)$, pk_H the hub's four one-time keys. As b does, the hub's E, its mode and K enter as private inputs, each authenticated by the hub's one-time signature under a key the statement fixes, and Script checks the same signatures in `respond`. Alice wins on any of:

- *return*: the hub signed b, and a return of b to c is anchored (P_1), in either mode;
- *claim*: the hub signed b and E in the plain leaf, and her claim holds: her notes in S's tree and unspent there, signed by their owners, distinct, S anchored with W on it, `request` in a block with W on it too and posting exactly this claim, b at most their sum;
- *failed counter*: the hub signed E and K in the countering leaf, and K is not a valid counter for her $\mathcal{D} \,\|\, E$.

Which key signed E says which leaf ran. A true counter leaves her no branch, and `respond_timeout` takes the contract once the deadline T_A passes. A countered claim loses the whole contract, as any false claim does; a careful wallet never makes one. A failed counter pays Alice whatever her own claim: a counter is always the hub's choice, and one that fails is one it chose to forge.

The hub's side. When a claim's `request` confirms, the hub reads the claim and searches: another of its contracts' requests signed by one of the owners, another contract's exit record, or a spend anchored by `request`'s block, its own return of at least b excluded. With a ground, it assembles the counter guest's input, waits until the blocks it needs are deep enough, proves and wraps the counter, and responds countering. With none, it asks the sequencer to refuse spends of the claimed nullifiers, records the exit on the L2 and responds plainly. An honest hub may fold an exit instead, having checked it natively.

What it guarantees. Not that every redemption consumes exactly one L2 entitlement whatever happens: a dead operator posts no exit records, so a note can be redeemed twice. The invariant is instead that no user loses to another or to the operator, since an honest claim can't be countered and every payment received stays good; that every payout not matched by value retired on the L2 is the operator's loss; and that an operator that stays live and follows the protocol can avoid that loss. The last fails within the anchor's latency until sender acknowledgment (Section 18.1): a dishonest Alice who pays Bob and requests before the payment's anchor confirms is paid, and Bob keeps the payment, at the hub's cost, once per contract.

16. What the exit costs

The proofs. Table 9 gives each statement's cycles, run on made-up inputs with the chain journal assumed (lichen), and its proving time. Given a valid counter, Alice's statement refuses

statement	cycles, segments	proving
a return's dispute, with the exit's guest	489k, 1	640 s and a 257 s wrap on the server (lichen)
a claim of one note	1.74M, 2	122 s on the laptop, lifted and joined (lichen)
a claim of eight notes	5.27M, 6	about 6 min on the laptop (est.)
the hub's counter, a spend	594k, 1	about a minute on the laptop (est.)
the hub's counter, an equivocation	309k, 1	30 s on the laptop; 306 s and a 252 s wrap on the server (lichen)
a failed counter	56.3M, 58	about an hour on the laptop, and a 134 s wrap; about ten hours on the server (est., from measured segments)

Table 9: The exit's statements: cycles, and proving times.

her, both natively and in the guest (lichen). The hub's proof is a small part of its window. The guest with the exit is 892 kB. The failed counter is Alice's worst case, the subject of Section 17.

On chain. On regtest under mainnet's relay policy (lichen): `request` 2.37 kvB with the largest claim, against 0.45 kvB without the exit; `respond` 4.48 kvB plain and 12.26 kvB countering (766 witness elements), against 0.76 kvB. So a refused true claim is about 15 kvB rather than 9.7, and an exit from a vanished hub, `request` and its timeout, about 2.9 kvB rather than 1.2 (est., from these sizes). The exit adds one pre-signed transaction, since `respond`'s two leaves are each pre-signed: 20 in all. None of it grows with misbehaviour.

Windows. The hub must find and prove a counter within its response window w_H: the contract's work on `request`'s block first, then the proof and the wrap. So w_H is 240 blocks on Mutinynet (two hours), 36 on signet (six hours) and 72 on testnet4. Alice's proof has no window: `assert` is due by an absolute deadline T_A, shortly before T_{close}.

Congestion. The rule of Table 5 applies to the hub's moves. Its windowed sum grows from 21.5 to about 25.2 kvB, or 33 kvB if every response counters, so a dispute of a week holds about 40,000 contracts (30,500) rather than 47,000 (est.). Alice's windowed move is unchanged. A wave of stale claims would make the hub prove many counters in one w_H; it is costly to mount, since each countered claimant loses a contract, and counters prove in parallel.

17. Alice's worst case

Why it is heavy. The failed-counter branch must show that a Groth16 proof does *not* verify. RISC Zero composes receipts cheaply, as the withdrawal does the chain receipt, but composition only shows that a receipt does verify; showing that one fails means running the verifier. K has

to be a Groth16 proof, not a RISC Zero receipt, because a `respond` leaf can bind only about 128 bytes of signed digits. So the guest verifies BN254 Groth16 in software, with RISC Zero's 256-bit accelerator: 56.3M cycles in 58 segments of 2^{20} with RISC Zero's fork of substrate-bn, 174M with arkworks, run with K a real proof's points (lichen). A profile of the 56.3M (lichen): the pairings 44%, of which Miller loops 10.7M, the final exponentiation 8.2M and precomputation 6.6M; checking that the verifying key's constant G2 points are in their subgroup 31%; the proof's own points' decompression and subgroup checks about 5.5M; the public input's scalar multiplications 8%.

What it takes. The branch's segments prove in about 53 s each on the laptop and take 11 s more to lift and join, so about an hour, then 134 s to wrap. On the 4-vCPU server it would be about ten hours. RISC Zero proves on the GPU with CUDA only; at the throughput it reports for an RTX 4090, about 1.4 MHz [16] (not checked by us), the segments take about a minute (est.).

Why it is tolerable. Three things soften it.

- *It is reached only by a forgery.* A hub that counters falsely loses the contract, so a rational hub never makes Alice run it.
- *It has days.* `assert`'s deadline is absolute, close to the contract's end; an hour, or ten, is no constraint.
- *Anyone can prove it.* Its inputs are on chain (E, K and the hub's signatures) or in the journal (c and $\mathcal{D}$), so it needs no secret of Alice's: a proving service, her tower or a friend can make it, and she signs the Groth16 proof she is handed after checking it.

But a deterrent counts only if it is credible, and an ordinary user on a phone cannot prove for an hour. In practice the branch presumes proving infrastructure that someone runs for her. Every branch already asks her for a proof and a wrap, minutes on a laptop; this one asks thirty times more, and that is where the design is weakest.

Alternatives. What we know of, and where each stands:

- *A cheaper verifier.* Skipping the subgroup checks of the verifying key's constants and precomputing its constant terms would bring the branch to about 31M cycles (est.): half an hour on the laptop. A new guest, so a new deployment.
- *The equivocation ground without a SNARK.* That ground needs only the owner's signature, so the hub could sign the evidence itself instead of a proof of it. It does not touch the worst case, which the hub can always choose, but it shrinks the countering `respond` and spares the hub a proof.
- *The hub signs the counter's facts, not a proof* (which note, how it was spent, at which height and root), and Alice refutes them from her own data, cheaply. Rejected: the operator can anchor a state it never proves, with her nullifier in it, and cite it. Against a state with no valid receipt Alice has nothing positive to show, whereas K proves that a valid one exists. It would also turn every withheld later state into a weapon against idle users, whom the design now protects.

- *A second BABE instance, mirrored.* The hub proves its counter in an instance of its own, with Alice as the garbler, and E, signed by her in `request`, authenticates its input as the hub's signature authenticates b in hers. Alice's statement loses the failed-counter branch; in a dispute she checks K natively, in milliseconds, and challenges a false one. The cost moves to every contract: Alice garbles at each opening and renewal (11 s of computation on the laptop, as the hub's setup is now) and sends the hub about 240 MiB, and a contested counter costs her a `challenge_assert` of about 20.7 kvB, as the hub pays now (est., from the measured setup and sizes). With contracts of months that is a modest price, and the one alternative that removes Alice's proving altogether.

18. Open questions

18.1. Data availability

The exit is safe exactly when Alice can read the anchored data she needs. Anchors are public; their contents are not. The sharpest form is the *consolidation attack*: Alice consolidates her notes into one, as her wallet recommends; the operator anchors the consolidation and withholds the new note's data. Her old notes are countered, the consolidation really spent them, and the new one has no kit. The same works on any spend that leaves her change. If nothing has been anchored since her kit, she is safe.

A counter wins the hub a contract, so withholding is harmless whenever a spend whose data Alice can't read can't be cited. Most remedies are rules of that kind [18]:

- *Sender acknowledgment*, the planned one. A spend can be cited only with its sender's signature that she holds its data, which her wallet gives once it has her kit at the anchored state containing the spend; until then the spend's outputs are locked, and only she may claim them. No trust and nothing on Bitcoin, at the price of two-phase finality: Bob's payment is final once the acknowledgment is anchored too. It also closes the race of Section 15, since the cutoff can then allow a grace the other way.
- *Publishing each anchor's inserted nullifiers and new commitments*, about 24 vB per one-in-two-out transaction: no trust, a cost per transaction on Bitcoin.
- *A data committee in the seal*: the seal's key a threshold key of the operator and a committee, each member signing an anchor only once it holds the batch's data. One honest signer suffices; the price is the committee's liveness for progress.
- *Accept the gap*: a user is exit-safe only while the operator serves data for her latest transactions.

18.2. The gadget

- The security argument of Section 5.5, written as a proof.
- A proof of robustness under cut-and-choose, against a hub that garbles wrongly. It is BABE's to give, but lichen's users rely on it.
- The security argument for the secret-sharing setup, with the shares and adaptor signatures in place of soldering.

- What Alice must download, about 240 MiB a contract: Duty-Free Bits' smaller encoding, and the kept instances' reuse across renewals.
- The anchors' own key, so that a tower can bump Alice's links above their floor.
- Whether BABE's authors, or GOAT's, see a problem with moving the amount into the witness this way.

18.3. The L2 and the exit

- The exit played on chain with real proofs, and the failed-counter branch proved whole, not only estimated from other inputs' segments.
- Alice's worst case: whether the mirrored instance, or something better, should replace the failed-counter branch (Section 17).
- Shielded notes' exits. A shielded note's authority is knowledge of nk inside the proof, with nothing to post or cite, so today a wallet exits its transparent notes only. Shielded addresses would gain a BIP340 owner key, bound into the nullifier, whose signature on E is posted like a transparent owner's; an exit would then reveal that key and link the wallet's exits, though not its spends.
- A tower for the exit. Alice's proof depends on which leaf the hub responds with, so unlike a return's she cannot sign it ahead; she must come back after `respond`, or entrust a key.
- A proof of the heaviest chain, or a counter-proof of a heavier one, in place of the work threshold W.
- Mass exit: every contract claiming by T_{close} against a hostile operator. The counter adds no games, since a failed counter pays Alice, but larger requests and a longer schedule, which favour few notes per claim, and so consolidation, which needs data availability.

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